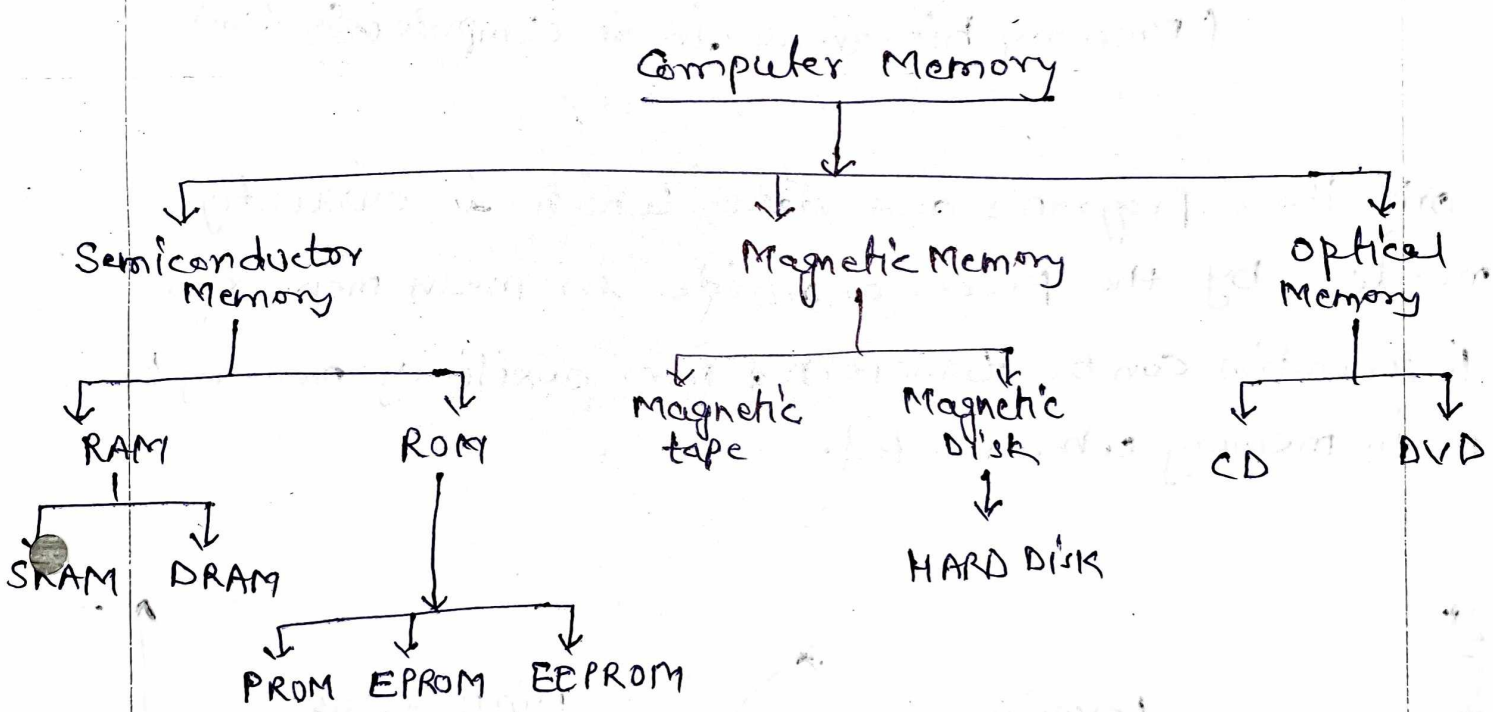
Unit - V

Memory classification:-

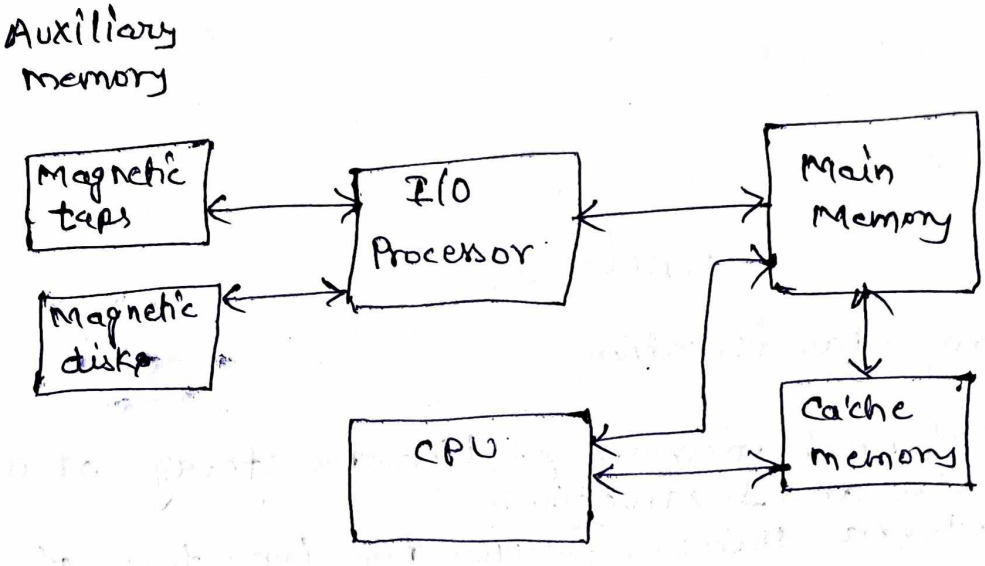
(i) Internal memory section for storage of active data and instructions.
(ii) External memory section for long term storage.



classification:-

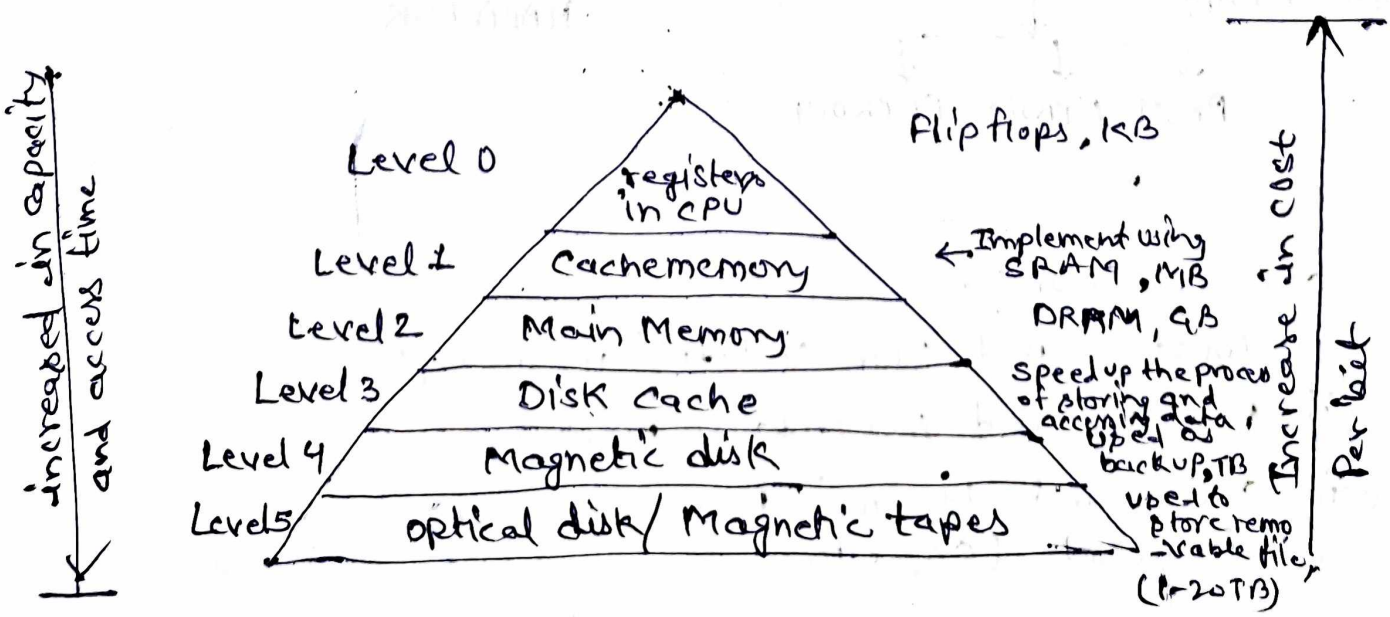
(i) Volatile and Non-Volatile memory
(ii) Primary and secondary memory
(iii) Static and Dynamic Memory
(Semiconductor) (Capacitor)

Memory Hierarchy :-



(Memory hierarchy in a Computer System)

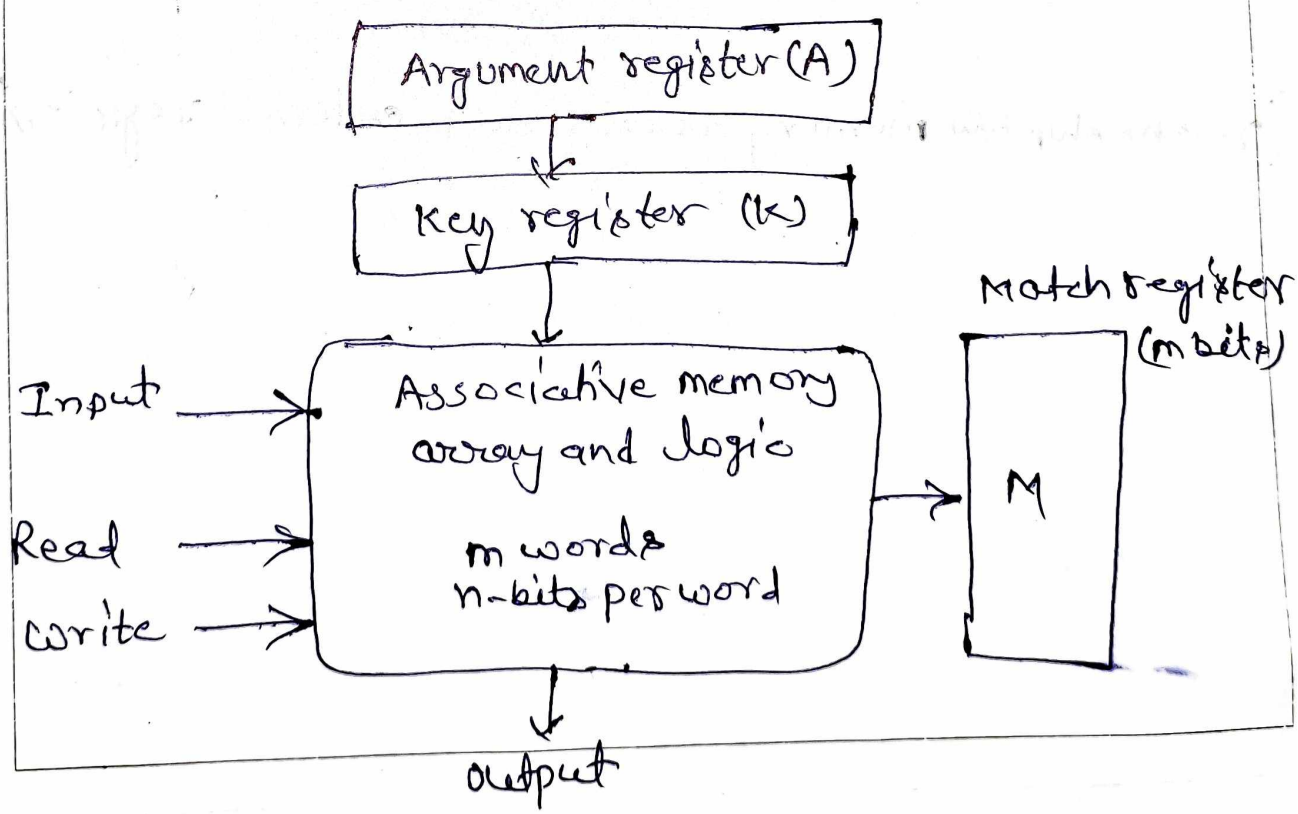
only those programs and data, which is currently needed by the processor, reside in main memory. Information can be transferred from auxiliary memory to main memory when needed.



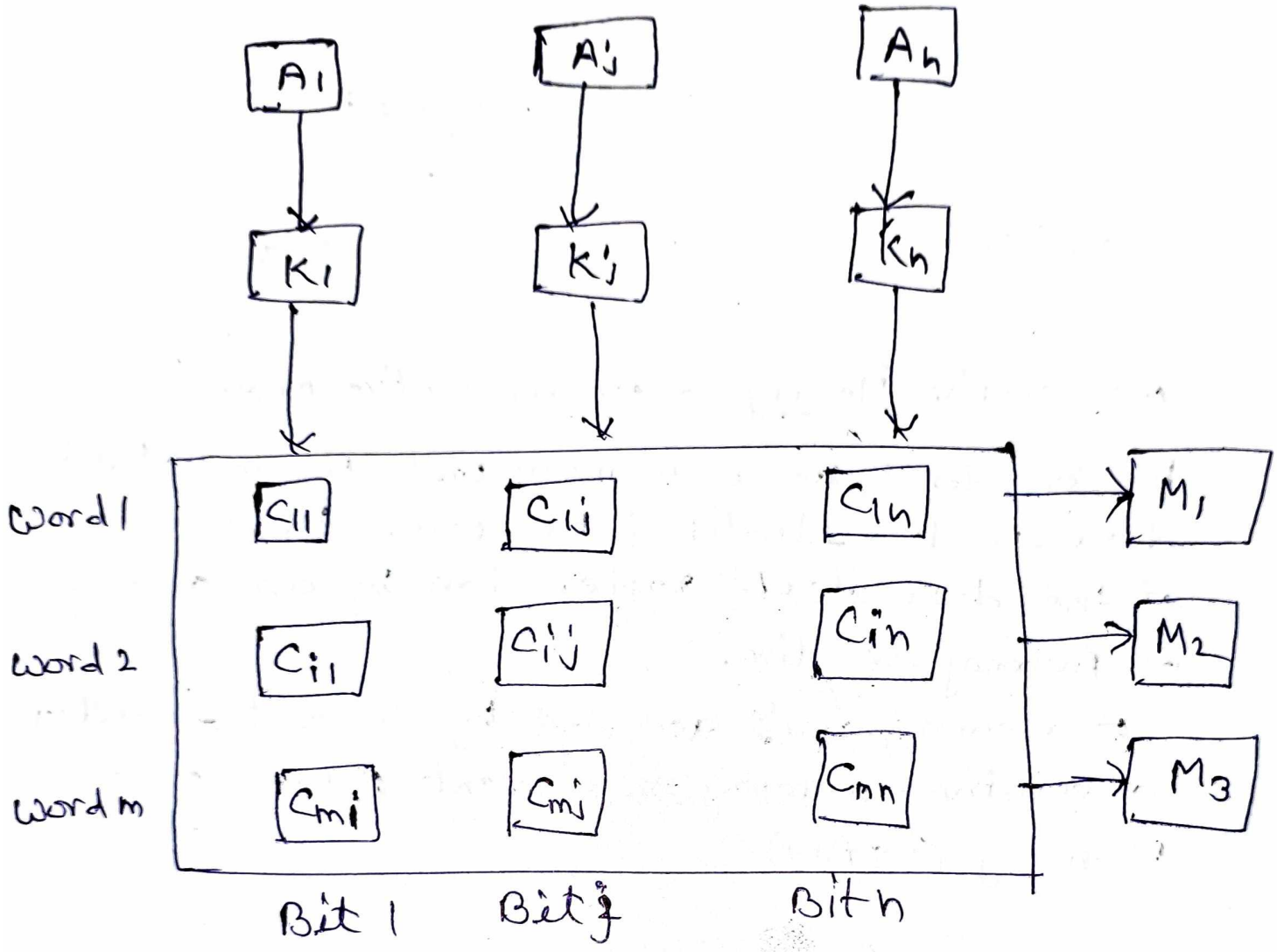
(Memory Heirarchy in term of access time and cost)

Associative Memory: → An associative memory can be considered as a memory unit whose stored data can be identified for access by the content of the data itself rather than by an address or memory location.

A memory unit accessed by content is called associative memory or Content Addressable Memory (CAM).



(Associative Memory)



(relationship b/w memory array and external registers)

Cache Memory :- Cache memory is a small sized type of volatile computer memory that provides high-speed data access to a processor and stores frequently used computer programs, applications and data. It is fastest memory in a computer, as is typically integrated onto the motherboard and directly embedded in the processor or main random access memory (RAM).

cache memory is costlier than main memory or disk memory but economical than CPU registers.

cache memory is an extremly fast memory type that acts as a buffer between RAM and the CPU. it holds frequently requested data and instructions so that they are immediately available to the CPU when needed

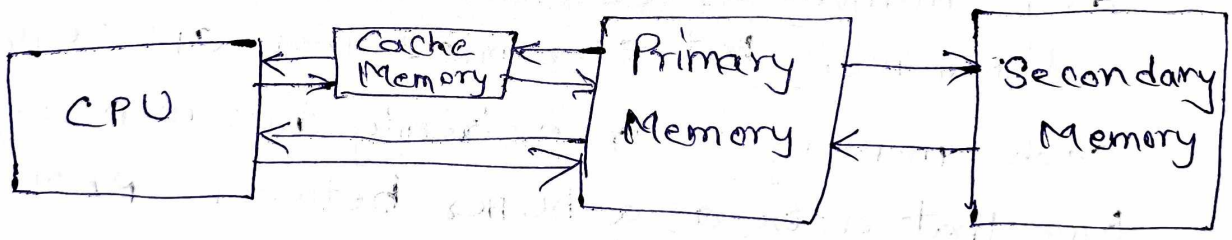
cache memory is used to reduce the average time to access data from the main memory.

The cache is a smaller and faster memory

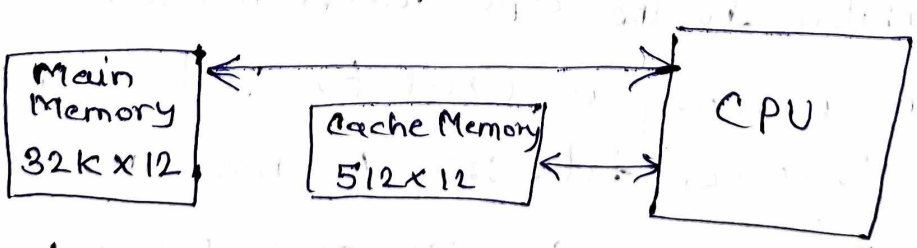
which stores copies of the data from frequently used main memory locations.

There are various different independent cache in a CPU, which stored instruction and data.

When the processor needs to access memory, the cache memory is checked. If the required word is found in cache memory, the word is delivered to the processor. If not, a block of main memory, consisting of same fixed number of words, is read into the cache and then word is transfered to the processor. In this way, some data is transferred to cache so that future references to memory, find the required words in the fast memory.



(Cache memory).



The main memory can store 32K words of 12 bits each. The cache is capable of storing 512 of these words at any given time.

Cache Performance :- The performance of the cache memory is measured in terms of a quantity called "hit ratio". If the desired word is found in cache memory, it is said to produce a hit. If the word is not found in cache it is referred as a "miss".

The ratio of the number of hit to the total Processor references to memory (hits + Misses) is defined as "hit ratio".

$$\text{Hit ratio} = \frac{\text{Hit}}{\text{Hit} + \text{Miss}}$$

High value of hit ratio is desirable to make the system fast. if the hit ratio is high enough so that most of the processor access the cache memory instead of main memory.

Therefore, average access time is closer to the access time of the fast cache memory

SCAN AnyScanner

Ex: Determine the average access time of a Computer system with a hit ratio of 0.95. The cache access time is 100 n sec and main memory access time is 1000 n sec.

Solⁿ Hit ratio = 0.95.

it means if process ~~reffer~~ referred to a word 100 times, found 95 times in cache memory and 5 times in main memory.

Therefore average access time is given as follows

$$T_{avg} = \frac{95 \times 100 + 5 \times 1000}{100} \text{ n sec} = 145 \text{ n sec.}$$

SCAN AnyScanner

Cache Mapping :- The transformation of data from main memory to cache memory is known as a "mapping processing".

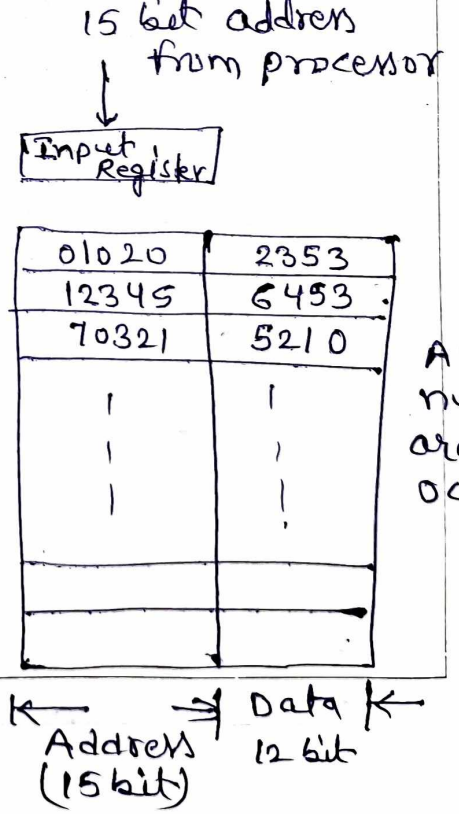
The mapping process plays an important role in reducing the search time in cache memory.

There are three types of the most common mapping processes.

(i) Associative mapping
(ii) Direct mapping
(iii) Set associative mapping

(i) Associative Mapping :-

Associative mapping cache stores both address and content of the memory word.



(Associative Mapping Cache)